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APPLICATION EXAMPLE

Redundant Open User Communication

LComRed Library

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1. Introduction

1.1. Overview

The Open User Communication (OUC) protocol based on TCP or ISO-on-TCP is used for simple Ethernet communication between controllers. However, the TCON module responsible for establishing the connection does not have the option of switching to an alternative connection if the connection fails.

In critical infrastructures or generally in applications with high availability requirements, failures caused by this cannot be tolerated.

The blocks of the "LComRed" library offer the following two functions as a solution for this:

- Monitoring of the connection
- Automatic switching to another connection path in case of disconnection

1.2. Principle of operation

The blocks of the LComRed library select a connection path from up to four connection paths as the active connection. For a fast connection diagnosis, a test connection is also established between the two communication partners and monitored with the help of test telegrams. If the monitoring detects a malfunction of the connection, the test and active connection are disconnected and a new active connection is searched for.

The blocks provide the user with a connection ID for the communication requests and a feedback as to whether this connection is currently available and active.

1.3. Components used

This application example consists of the following libraries:

Component	File name	Note
LComRed for TIA Portal V20	LComRed	
LComRed Classic for STEP 7 ≥ Version 5.7	LComRedClassic	

NOTE

All components of both libraries are freely available as sources in the article. These can be used to integrate the modules into earlier TIA Portal or STEP 7 versions than those specified here.

LComRed

The LComRed library comprises the following standardized blocks. The specified PLC data types are used internally in the function blocks and copied automatically to the PLC program as required by inserting one of the LComRed blocks.

Component	Type	Note
LComRed_ISOOnTCP	Function block	Red. communication via ISO-on-TCP
LComRed_TCP	Function block	Red. communication via TCP (optional encrypted)
LComRed_ISOOnTCP_Classic	Function block	Red. communication via ISO-on-TCP to S7-300/400
LComRed_typeTConHandlingRFC	PLC data type	Internal data type
LComRed_typeTConHandlingV4	PLC data type	Internal data type
LComRed_typeTConRedTestTelegramm	PLC data type	Internal data type

Component	Type	Note
LComRed_typeTConSecureConnection	PLC data type	Internal data type
LComRed_typeTDisconHandling	PLC data type	Internal data type
LComRed_typeTRcvHandling	PLC data type	Internal data type
LComRed_typeTSendHandling	PLC data type	Internal data type
typeDiagnostics	PLC data type	Internal data type

The LComRed library can be used in TIA Portal V20 on the following hardware families:

Component	Article number	Version	Note
S7-1200	6ES7 21.-.....-....	≥ 4.4	S7-1200
S7-1200 G2	6ES7 21.-.....-....	≥ 1.0	S7-1200 G2
S7-1500	6ES7 51.-.....-....	≥ 2.5	S7-1500
S7-1500R/H/HF	6ES7 51.-.....-....	≥ 3.1	Redundant S7-1500

NOTE

The mentioned restrictions regarding the firmware versions of the individual hardware families mainly refer to the LComRed_TCP block and the option of encrypted connections. The device uses a specific data structure (TCON_IP_SEC) internally for this purpose, which is only available from the versions mentioned.

The other function blocks can also be used on previous versions.

LComRed Classic

The STEP 7 Library LComRedClassic contains the following block:

Component	Type	Note
LComRed_ISOonTCP	Function block	Red. communication via ISO-on-TCP

The STEP 7 Library LComRedClassic can be used on the following hardware or the following hardware families:

Component	Article number	Version	Note
S7-300	6ES7 31.-.....-OAB0	≥ 3.2	
S7-400	6ES7 41.-.....-OAB0	≥ 6.0	

2. Engineering

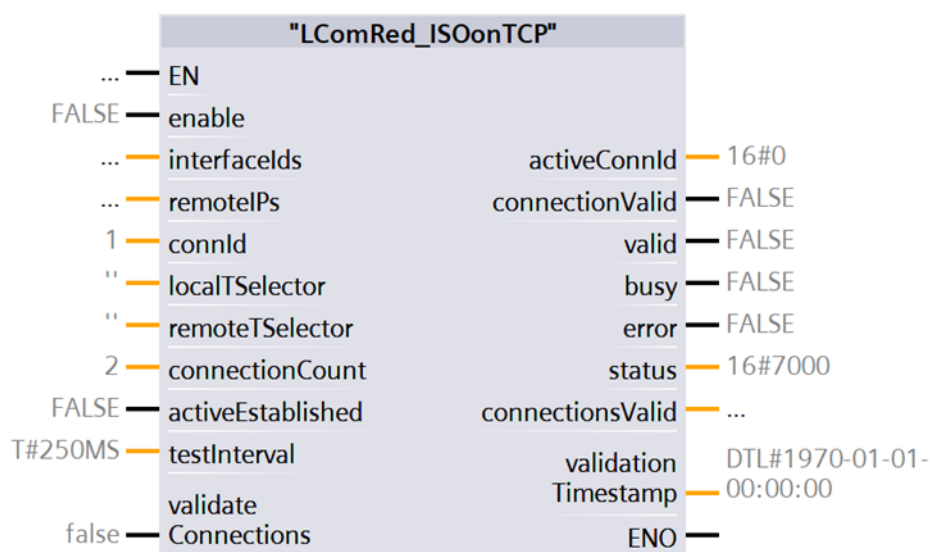
2.1. Interface description

The LComRed blocks are based on the block TCON used internally in terms of parameterization. With this block, the connection description is transferred via the input parameter "CONNECT".

In order to reduce the effort required to parameterize the LComRed blocks, some parameters of the input parameter "CONNECT" have been combined. It is not necessary to define a separate data set for each of the up to four connections. Apart from the IPs of the communication partner and the HW_IDs of the local communication interfaces, the remaining values are based on a start value or are determined based on this.

However, this simplification of operation cannot be fully applied to the LComRed_Classic blocks for communication between S7-300/400 and S7-1200/1500. On the S7-300/400 controller family side, the "CONNECT" data set contains information about the interface to be used and its position in the rack, which prevents generation on the basis of initial values.

2.1.1. LComRed_ISOonTCP (S7-1200/1500)



Inputs of the LComRed_ISOonTCP

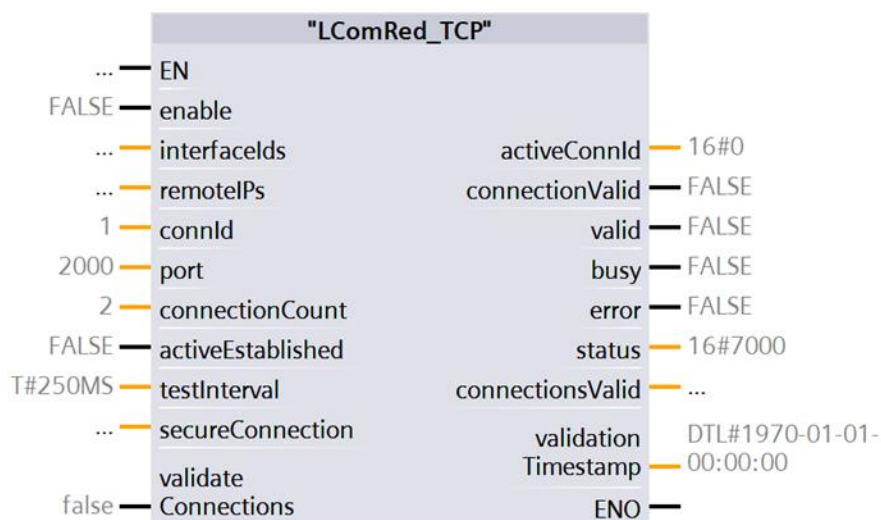
Input	Type	Start value	Note
enable	Bool	False	Activates the connection establishment and monitoring
interfacelds	Array[1..4] of HW_ANY	-	Hardware IDs of the local interfaces
remotelPs	Array[1..4] of IP_V4	-	IP addresses of the partner endpoint
connId	UInt	1	Start value for connection reference
localTSelector	String[14]	-	TSelector start value of the local connection partner
remoteTSelector	String[14]	-	TSelector start value of the removed connection partner
connectionCount	UInt	2	Number of connection paths (min = 2, max = 4)
activeEstablished	Bool	False	ID for type of connection establishment FALSE: Passive connection establishment TRUE: Active connection establishment
testInterval	Time	T#250ms	Test telegram cycle

Input	Type	Start value	Note
validateConnections	Bool	-	True: Test of all parameterized connections

Outputs of the LComRed_ISOonTCP

Output	Type	Note
activeConnId	CONN_OUC	Reference to the active communication connection
connectionValid	Bool	Status of the active communication connection
valid	Bool	All output data is valid
busy	Bool	The module is active
error	Bool	Error status, refer to status for cause
status	Word	Status of the instruction
connectionsValid	Array[1..4] of Bool	Connection test feedback – index corresponds to the connection, True = connection could be established successfully
validationTimestamp	DTL	Date and time of the last connection test

2.1.2. LComRed_TCP



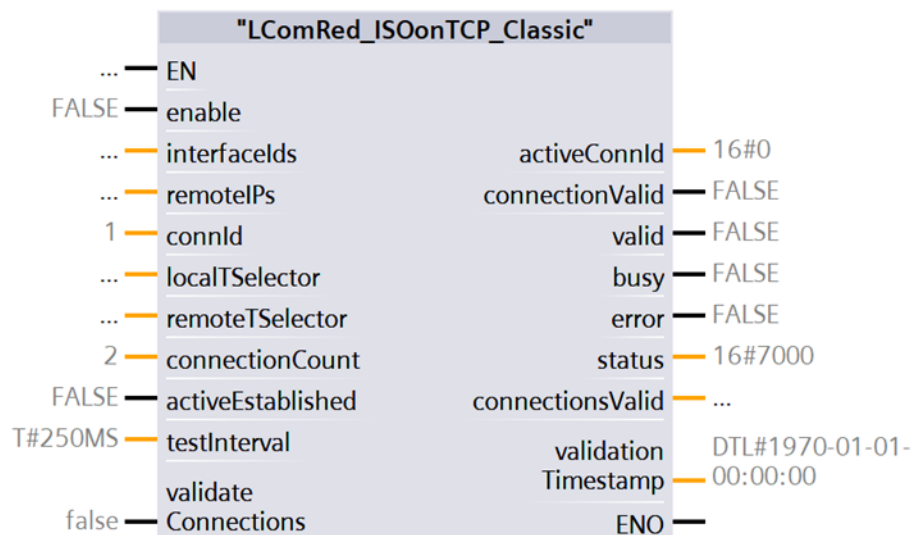
Inputs of the LComRed_TCP

Input	Type	Start value	Note
enable	Bool	False	Activates the connection establishment and monitoring
interfacelds	Array[1..4] of HW_ANY	-	Hardware IDs of the local interfaces
remotelPs	Array[1..4] of IP_V4	-	IP addresses of the partner endpoint
connId	UInt	1	Start value for connection reference
port	UInt	2000	Start value port address of the passive connection partner
connectionCount	UInt	2	Number of connection paths (min = 2, max = 4)
activeEstablished	Bool	False	ID for type of connection establishment FALSE: Passive connection establishment TRUE: Active connection establishment
testInterval	Time	T#250ms	Test telegram cycle
secureConnection	LComRed_typeTConSecureConnection	-	Parameters for the optional encryption of the connections
validateConnections	Bool	-	True: Test of all parameterized connections

Outputs of the LComRed_TCP

Output	Type	Note
activeConnId	CONN_OUC	Reference to the active communication connection
connectionValid	Bool	Status of the active communication connection
valid	Bool	All output data is valid
busy	Bool	The module is active
error	Bool	Error status, refer to status for cause
status	Word	Status of the instruction
connectionsValid	Array[1..4] of Bool	Connection test feedback – index corresponds to the connection, True = connection could be established successfully
validationTimestamp	DTL	Date and time of the last connection test

2.1.3. LComRed_ISOonTCP_Classic (S7-1200/1500)



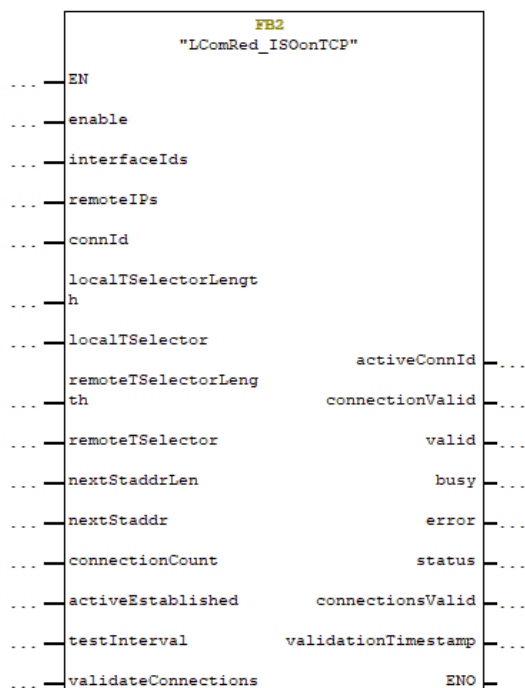
Inputs of the LComRed_ISOonTCP_Classic

Input	Type	Start value	Note
enable	Bool	False	Activates the connection establishment and monitoring
interfacelds	Array[1..4] of HW_ANY	-	Hardware IDs of the local interfaces
remotelPs	Array[1..4] of IP_V4	-	IP addresses of the partner endpoint
connId	UInt	1	Start value for connection reference
localTSelector	Array[1..4] of TSelector	-	TSelectors of the local connection partner
remoteTSelector	Array[1..4] of TSelector	-	TSelectors of the remote connection partner
connectionCount	UInt	2	Number of connection paths (min = 2, max = 4)
activeEstablished	Bool	False	ID for type of connection establishment FALSE: Passive connection establishment TRUE: Active connection establishment
testInterval	Time	T#250ms	Test telegram cycle
validateConnections	Bool	-	True: Test of all parameterized connections

Outputs of the LComRed_ISOonTCP_Classic

Output	Type	Note
activeConnId	CONN_OUC	Reference to the active communication connection
connectionValid	Bool	Status of the active communication connection
valid	Bool	All output data is valid
busy	Bool	The module is active
error	Bool	Error status, refer to status for cause
status	Word	Status of the instruction
connectionsValid	Array[1..4] of Bool	Connection test feedback – index corresponds to the connection, True = connection could be established successfully
validationTimestamp	DTL	Date and time of the last connection test

2.1.4. LComRed_ISOonTCP (S7-300/400)



Inputs of the LComRed_ISOonTCP

Input	Type	Start value	Note
enable	Bool	False	Activates the connection establishment and monitoring
interfacelds	Array[1..4] of byte	-	Corresponds to the local_device_id of the TCON
remoteIPs	Array[1..4, 1..4] of byte	-	IP addresses of the partner endpoint
connId	Int	1	Start value for connection reference
localTSelectorLength	Array[1..4] of byte	-	Length of the TSelectors of the local connection partner
localTSelector	Array[1..4, 1..16] of byte	-	TSelectors of the local connection partner
remoteTSelectorLength	Array[1..4] of byte	-	Length of the TSelectors of the remote connection partner
remoteTSelector	Array[1..4, 1..16] of byte	-	TSelectors of the remote connection partner
nextStaddrLen	Array[1..4] of byte	-	Length of the nextStaddrLen parameter
nextStaddr	Array[1..4, 1..6] of byte	-	Rack and slot for the communications interface
connectionCount	Int	2	Number of connection paths (min = 2, max = 4)
activeEstablished	Bool	False	ID for type of connection establishment FALSE: Passive connection establishment TRUE: Active connection establishment
testInterval	Time	T#250ms	Test telegram cycle
validateConnections	Bool	-	True: Test of all parameterized connections

Outputs of the LComRed_ISOonTCP

Output	Type	Note
activeConnId	Word	Reference to the active communication connection
connectionValid	Bool	Status of the active communication connection
valid	Bool	All output data is valid
busy	Bool	The module is active
error	Bool	Error status, refer to status for cause
status	Word	Status of the instruction
connectionsValid	Array[1..4] of Bool	Connection test feedback – index corresponds to the connection, True = connection could be established successfully
validationTimestamp	DATE_AND_TIME	Date and time of the last connection test

2.2. Integration in the user project

2.2.1. LComRed

From version 1.2 onwards, the function blocks of the LComRed library are unencrypted, stored in a typed form and can be integrated into the user program via drag & drop. PLC data types used internally are automatically adopted.

A functioning connection between active and passive connection partners can only be established if the correct block is used respectively on both sides.

The following block combinations are supported:

Communication partner 1		Communication partner 2
S7-1200/1500 "LComRed_TCP"	↔	S7-1200/1500 "LComRed_TCP"
S7-1200/1500 "LComRed_ISOonTCP"	↔	S7-1200/1500 "LComRed_ISOonTCP"
S7-1200/1500 "LComRed_ISOonTCP_Classic"	↔	S7-300/400 "LComRed_ISOonTCP" (LComRed Classic)

The "Classic" block of the S7-1200/1500 controllers is provided for redundant communication with controllers of the S7-300/400 family.

2.2.2. LComRedClassic

The LComRedClassic Library for utilization in STEP 7 as of Version 5.7 includes the encoded block "LComRed_ISOonTCP". This calls the following blocks or UDT internally, which must be available in the program block folder:

- TSEND [FB63]
- TRCV [FB64]
- TCON [FB65]
- TDISCON [FB66]
- TCON_PAR [UDT65]

These blocks can be found in the "Standard Library" in the "Communication Blocks" folder.

As well as the system module or the system function:

- TON [SFB4]
- READ_CLK [SFC1]

Note

The LComRed_ISOonTCP block uses "Array" as a data type for some of the inputs. However, this data type is not supported in the Continuous Function Chart (CFC) language. Consequently, the LComRed_ISOonTCP cannot be used directly in a CFC.

2.3. Operation

An LComRed block can be called either in a cyclic or in a cyclic interrupt OB. Before the "enable" input is activated, you must have provided the following parameters correctly and in their entirety:

- interfacedIds
- remoteIPs
- connID
- remoteTSelector
- remoteTSelectorLength (only for S7-300/400)
- localTSelector
- localTSelectorLength (only for S7-300/400)
- nextStaddr & nextStaddrLen (only for S7-300/400)
- connectionCount
- activeEstablished
- testInterval

With activation of the "enable" input on the active and passive communications side, the currently established connection is provided at the "activeConnId" output for utilization at a corresponding TSEND/TRCV if the parameterization is correct. The connection status is conveyed via the "connectionValid" output.

Details regarding the individual parameters are explained in the following chapter.

2.4. Parameter details

2.4.1. Input enable

An LComRed block only functions as long as the "enable" input is set to "True". Changes to the other input parameters of the block are ignored with this setting. In order that this can be adopted, a rising edge is required at the "enable" input.

With falling edge at the "enable" input, all connections are deactivated internally via TDISCON. Should the input become enabled again via the user program in the meantime, this will only be executed following a delay to allow complete deactivation of the connection. The "enable" input, however, does not have to be triggered again in this regard.

2.4.2. Inputs remotelPs & interfacelds

Up to four connection data records are transferred to an LComRed block on the active and the passive communication side. These include for each connection:

- IP address of the communication partner for each connection
 {remotelPs: Array [1..4] of IP_V4}
 or
 {remotelPs: Array [1..4, 1..4] of byte}
- The interface identifier of the local communications interface to be used for each connection. For S7-1200/1500, this would entail the HW ID of the interface being used, and for S7-300/400 the local_device_id.
 {interfacelds : Array [1..4] of HW_ID}
 or
 {interfacelds : Array [1..4] of byte}

It is recommended that the individual connection paths are visualized and "remotelPs" and "interfacelds" entered in order that the correct data for the respective connection can be provided on both sides. Examples of correct parameterization can be obtained in chapter [3.3](#).

The number of data records which must be filled depends on the "connectionCount" input. If this is parameterized to "2", for example, "remotelPs[1] and [2]", as well as "interfacelds[1] and [2]" must be filled with valid values on the active and passive side respectively. The remaining elements can be left blank as they are not evaluated by an LComRed block.

With a "connectionCount" of "4", both parameters must be filled with values in their entirety.

2.4.3. Input connId

Connection IDs are used as a reference to a connection. The first ID to be used by the LComRed block is transferred to the block via the "connId" parameter. The IDs of the block being used, and consequently disabled for other OUC communication paths, are yielded together with the specification regarding number of connection paths ("connectionCount"). Additionally, a further ID is used for the connection diagnostics to achieve a faster connection switchover.

If the "connId = 10" and a connection number of "2" is transferred for the block, for example, the block disables the IDs "10" and "11", as well as "12" for a test connection. 3 connection resources are therefore required.

With a "connId = 112" and four connections, the IDs "112", "113", "114" and "115", as well as "116" for the test connection are disabled, which in turn requires 5 connection resources.

The "connId" does not have to be identical for the active and passive side, as these are only used for PLC-internal referencing of a connection. No adjustment of the IDs is made between the two communication partners.

If the block is active ("enable = true") and has a valid active connection, two connection resources are required, one for the active connection and one for the internal test connection.

Upon activation of the LComRed block, for the connection switchover and when using the "validateConnections" function, the LComRed block briefly requires the full number of resources for the possible connection paths + test connection:

- "connectionCount" = 2 → required resources 3
- "connectionCount" = 3 → required resources 4
- "connectionCount" = 4 → required resources 5

If you are using multiple LComRed instances for each PLC/system, these must be considered with regard to the connection resources available for the system.

2.4.4. Inputs local & remote TSelector for LComRed_ISOonTCP (S7-1200/1500)

The ISOonTCP communication requires a Transportation Service Access Point (TSAP) for Local and Remote respectively for each connection. To ensure that you will not have to specify two TSAPs ("local & remote TSelector") respectively for each data record, only one TSAP (local & remote) is transferred respectively at the LComRed_ISOonTCP. The TSAP must feature a length of at least "3", however, may not exceed "14". The limitation to "14" in contrast to the familiar "16" for the standard TCON is due to the internal processing of the LComRed_ISOonTCP. For each of the four maximum possible connections, the TSAP is extended by "_x" (x = 1..4), or "_T" for the test connection.

For example, the input parameter given as a simple string...

- "localTSelector.TSel" = "ISOonTCP"

... becomes the following in the internal processing with a "connectionCount = 4":

- Connection 1: "localTSelector.TSel" = **ISOonTCP_1** (49.53.4F.6F.6E.54.43.50.5F.31)
- Connection 2: "localTSelector.TSel" = **ISOonTCP_2** (49.53.4F.6F.6E.54.43.50.5F.32)
- Connection 3: "localTSelector.TSel" = **ISOonTCP_3** (49.53.4F.6F.6E.54.43.50.5F.33)
- Connection 4: "localTSelector.TSel" = **ISOonTCP_4** (49.53.4F.6F.6E.54.43.50.5F.34)
- Test connection "localTSelector.TSel" = **ISOonTCP_T** (49.53.4F.6F.6E.54.43.50.5F.54)

thus becomes "localTSelector.TSelLength" = 10 for all connections.

2.4.5. Inputs local & remote TSelector for the Classic communication

For communication between S7-1200/1500 (LComRed_ISOonTCP_Classic) and S7-300/400 (LComRed_ISOonTCP), an individual and unambiguous local and remote TSAP must be specified for each of the possible 4 connections.

Each TSAP must feature a length of at least "3", however, may not exceed "14". The limitation to "14" in contrast to the familiar "16" for the standard TCON is due to the internal processing of the blocks. If one of these possible four connections becomes the active connection, the test connection takes its TSAPs as a basis and supplements each with "_T".

2.4.6. Inputs nextStaddr & nextStaddrLen (S7-300/400)

The LComRed blocks are based on the TCON blocks to establish the connection. The TCON versions of the S7-300/400 revert to the UDT 65 "TCON_PAR" connection description via the "CONNECT" input for transfer of the connection parameters. The slot and the rack number of the communications processor (CP) being used must be specified in this description in accordance with the "local_device_id" with the "nextStaddr".

Communication via a CP443-1 for an S7-400H, for example, thus requires a "nextStaddr[1]" = 16#24 and "nextStaddrLen" = 16#01 in rack 1 at slot 4.

2.4.7. Input port for LComRed_TCP

For the "port" input, the port number for the first connection set is transferred to the LComRed_TCP block. The smallest possible value is "2000" and is used for the active partner as "remotePort" for the first connection. Every further connection uses an incremental value internally. The port specifications must correlate on both the active and passive side. The "localPort" of the active partner is selected dynamically from the interface and therefore does not have to be prescribed.

2.4.8. Input connectionCount

LComRed blocks support two to a maximum four connections, between which switchovers can be performed.

2.4.9. Input activeEstablished

With the "activeEstablished" input, it is specified whether this involves the active or passive connection partner, as with the Standard TCON.

One side must be the active communication partner, and consequently the other side must be the passive communication partner.

2.4.10. Input testInterval

With the "testInterval" input, the interval at which the active connection partner is to send test telegrams via the test channel is specified with a time value. In this regard, all other internal monitoring times of the block are LComRed. It should be considered at this point that the smaller the value, the greater the switchover time is reduced in the event of an error, but the network and communications loads of the PLCs are increased by the telegrams.

In addition, it should be considered that the value must be greater than the total cycle time. Factor 5 is recommended. Thus, if the block is called with a cycle of 40 ms, the "testInterval" input should be set to "T#200ms". As the same value must be used on the active and passive side, the cycle time of the communication partner with the higher cycle time applies.

If the time selected is too small, the connection will be unstable. This will be noticeable as the "connectionValid" output repeatedly switches to "false", but the same connection is always established. The connection itself is therefore not disrupted, rather neither of the connection partners manage to successfully exchange a test telegram within the monitoring period prescribed.

For the "1513R" and "1515R" systems in particular, the LComRed block has a major influence on the total cycle time due to the synchronization via PROFINET. This should be monitored by the RT_INFO block with "Mode 25". This mode indicates the actual cycle (cyclic OBs, interrupt OBs, communication, synchronization). The "testInterval" input must be adapted accordingly.

2.4.11. Encrypted connection

The LComRed_TCP block supports encrypted communication between a TLS client (active connection partner) and a TLS server (passive connection partner) from version 1.2 onwards.

The settings correspond to the parameters of the TCON_IP_V4_SEC data type of the TCON device.

- ActivateSecureConn
- TLSServerReqClientCert
- ExtTLSCapabilites
- TLSServerCertRed

Please refer to the corresponding TCON documentation on how to parameterize the parameters for a secure connection ([13](#)).

2.4.12. Connection test via the validateConnections input

The input "validateConnections" triggers a connection test internally via a positive edge. The partner system is informed via the test channel that all passive connections should be opened and tested. The test can be triggered on the active as well as on the passive communication partner.

The result can be found in the field "connectionsValid". The index corresponds to the connection. If the boolean value is true, the corresponding connection could be successfully established and disconnected during the test.

The timestamp of the last successful test is output in the "validationTimestamp" output



Risk of complete loss of communication

The LComRed modules do not check whether all connections can be established when activated in the first step. The connection that was the first to report a successful connection is always made available as an active connection.

Immediately afterwards, it automatically checks whether all passive connections are available.

This connection check should be triggered regularly via the "validateConnections" input and evaluated via "connectionsValid".

2.4.13. Output activeConnId

At the output, the block with the "activeConnId" supplies the value of the OUC connection currently established which can be used for TSEND/TRCV. The ID remains unstable throughout the process, i.e. it switches following connection errors to a range between

`"connId"` and `"connId + connectionCount - 1"`

For example `"connId = 200"`, `"connectionCount = 3"` → possible `"activeConnIds = 200, 201 or 202"`.

If data loss is also to be prevented during a switchover, buffering of the data must be additionally implemented in the user program.

2.4.14. Output connectionValid

The "connectionValid" output specifies whether the connection is successfully tested. This value specifies whether a test telegram was transferred once completely from the active to the passive communication partner and back after establishing the test connection. If the block cannot find a connection, the "activeConnId" retains the "connId" input parameter value and the "connectionValid" output remains "false".

As long as the "enable" input is "true" and the "Error" is "false", the LComRed block will continuously try to find an active connection.

2.4.15. Outputs valid, busy, error & status

The output "valid" indicates whether the outputs of the device are valid and can be used in the further program.

The output "busy" indicates that the function module is active and working.

The "error" output indicates whether there is an error. The necessary details are provided by the "status" output.

As long as an error is active on the LComRed device, no connection is established.

2.5. Error handling

Error	Status (W#16#...)	Explanation
0	0000	Connection successfully established
0	7000	No job active – enable = false
0	7001	Connection is being established
0	7003	Active connection down
0	7004	Test connection is being established
0	7005	Test connection established
0	7006	Connection test active
1	8200	connId outside the valid range
1	8201	connectionCount outside of the valid range (min = 2, max = 4)
1	8202	port outside the valid range
1	8203	testInterval must be at least 50 ms
1	8204	Length of the localTSelector is too short (min = 3) or too long (max = 14)
1	8205	Length of the remoteTSelector is too short (min = 3) or too long (max = 14)
1	82x4	Classic communication only - Length of the localTSelector too short (min = 3) or too long (max = 14) – x identifies the connection
1	82x5	Classic communication only - Length of the remoteTSelector too short (min = 3) or too long (max = 14) – x identifies the connection

3. Useful Information

3.1. Details of principle of operation

Familiarity with the internal functioning is not necessary for utilization of the LComRed blocks. This chapter therefore concentrates on interested parties or developers of communication drivers, e.g. for utilization on PCs or external PLCs.

Internally, the LComRed blocks function essentially in two phases which are explained in the following sections.

Phase 1 – Searching for a connection

With activation of an LComRed block via rising edge at the "enable" input, following a successful parameter check one TCON instance respectively is activated in accordance with the number of connections. With four connections, for example, four TCON instances accordingly are initiated on the active and passive side. An LComRed waits until one of the instances confirms that a connection has been successfully established. If this concerns more than one connection within a cycle, the connection with the lower number is preferred.

This connection becomes the active connection and the "activeConnId" output is updated accordingly. This OUC ID can be used for the communication via TSEND/TRCV.

Phase 2 – Validating the active connection

Upon conclusion of Phase 1, the "connectionValid" output is still "false", however; the connection is established but has not yet been validated.

Instead, an additional TCON establishes a test connection on the active and passive side. The data of the active connection ("HW_ID" or "local_device_id", "remoteIPs", ...) is used as a basis for the Connect parameters. Only the "connId" and "ports" or TSAPs are determined individually.

Test telegrams are exchanged between active and passive communication partners via the test connection. This is initiated from the active side in the cycle of the "testInterval" input.

If a test telegram is not received by the partner within the "testInterval" or is received with incorrect values, the test connection and the active connection are disconnected and the "connectionValid" output is set to "false". The LComRed block switches immediately to Phase 1 once again and searches for a new connection.

Structure of the test telegram

Element	Type	Note
key	DWORD	Key code for the respective connection
telegramNumber	INT	Numerical value of telegram
telegramConfirmed	INT	Copy of numerical value as confirmation
testInterval	TIME	Set testInterval on active side

"key" of the test telegram

This key value is calculated on the active side as follows:

$$\text{key} = a - b - c$$

With the following elements:

1. A constant (a) depending on the connection type

- ISOonTCP → a = 16#FFFB_BF3C
- TCP → a = 16#FFFA_A56C

2. A connection-specific value (b)

- ISOonTCP → b = checksum of the "localTSelector" input of the active communication partner
- TCP → b = the value of the input "port"

3. Number of connections (c) in the range from 1 to the value at the "connectionCount" input

The value of the "key" element is checked on the passive side and is copied in the confirmation telegram decremented by "1".

Upon receipt, the active side checks whether the value of the "key" element which it sent is greater than that received in the confirmation by precisely 1.

"telegramNumber" and "telegramConfirmed" of the test telegram

The active side sends a value between "1" and "32767" in the "telegramNumber". Upon reaching "32767", or each time the test connection is established, i.e. when the LComRed block is activated or a connection switchover occurs, this numerical value is reset to "1".

The value of the "telegramConfirmed" remains "0".

The passive communication partner copies the value of the "telegramNumber" to "telegramNumber" and "telegramConfirmed" in the confirmation telegram.

The active side checks whether "telegramNumber" and "telegramConfirmed" are identical in the confirmation telegram and whether the "telegramNumber" sent correlates with that received in the confirmation telegram.

"testInterval" of the test telegram

The active communication partner copies the value of the "testInterval" input for the LComRed block at this point.

The passive side updates the internally applied "testInterval" to this value after receiving the first test telegram once the test connection has been established.

3.2. Test of connection

Version 1.2 of the LComRed function modules supports a connection test. On request, this checks whether the passive connections would be available if the active connection failed.

For this purpose, after activation on the side of the active or passive connection partner, a certain value in the "key" is transmitted via the test connection to inform the partner that the passive connections should be opened.

The value for "key" is "16#FFFC_AAAA"

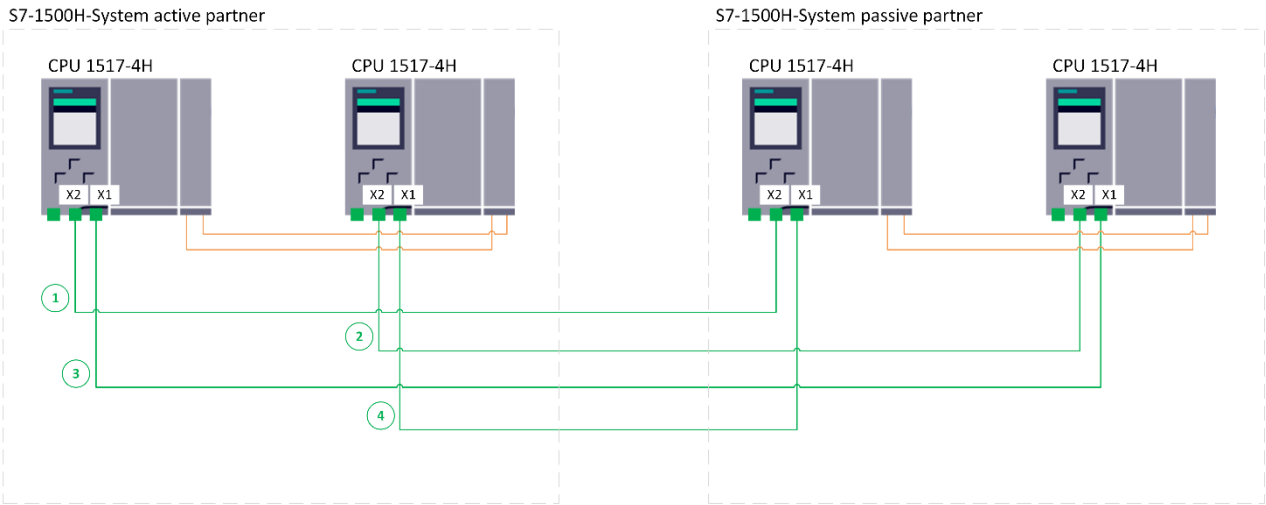
The partner responds to this with the same "key" to confirm the request. Both communication partners then open the unused connections for 5 times the time of the test interval.

The result is output via the "connectionsValid" field together with the timestamp ("validationTimestamp").

Especially during commissioning, this function can be used to evaluate whether all connection parameters have been set correctly.

3.3. Examples

3.3.1. LComRed_TCP - Redundant connection between two S7-1500H systems



Device properties active connection partner

Parameters	Value	Note
PLC	CPU 1517H-4 PN	6ES7517-4HQ10-0AB0
IP address interface X1	PLC1: 192.168.0.20 PLC2: 192.168.0.21	
IP address interface X2	PLC1: 192.168.1.20 PLC2: 192.168.1.21	
Minimum cycle time	1 ms	
Calling organization block	OB30	Cycle clock: 10 ms
LComRed block	LComRed_TCP	From LComRed

Device properties passive connection partner

Parameters	Value	Note
PLC	CPU 1517H-4 PN	6ES7517-4HQ10-0AB0
IP address interface X1	PLC1: 192.168.0.30 PLC2: 192.168.0.31	
IP address interface X2	PLC1: 192.168.1.30 PLC2: 192.168.1.31	
Minimum cycle time	1 ms	
Calling organization block	OB30	Cycle clock: 10 ms
LComRed block	LComRed_TCP	From LComRed

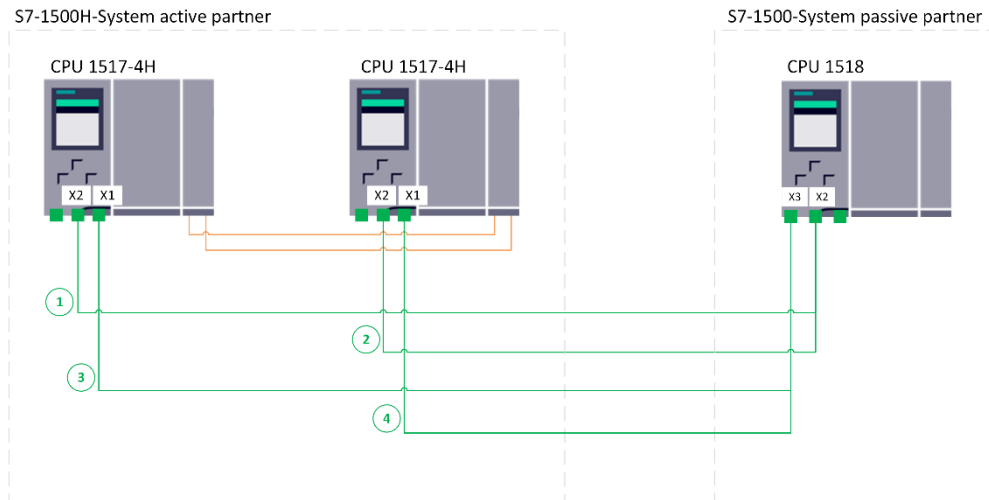
LComRed_TCP input parameter active connection partner

Parameters	Value	Note
interfacelds	interfacelds[1] := Local1~PROFINET-interface_2 interfacelds[2] := Local2~PROFINET-interface_2 interfacelds[3] := Local1~PROFINET-interface_1 interfacelds[4] := Local2~PROFINET-interface_1	PLC1.X2 PLC2.X2 PLC1.X1 PLC2.X1
remotelPs	remotelPs[1].ADDR[1] := 192 remotelPs[1].ADDR[2] := 168 remotelPs[1].ADDR[3] := 1 remotelPs[1].ADDR[4] := 30 Corresponding to remotelPs[2] := 192.168.1.31 remotelPs[3] := 192.168.0.31 remotelPs[4] := 192.168.0.30	
connId	1	
port	2000	
connectionCount	4	
activeEstablished	True	
testInterval	T#50ms	

LComRed_TCP input parameter passive connection partner

Parameters	Value	Note
interfacelds	interfacelds[1] := Local1~PROFINET-interface_2 interfacelds[2] := Local2~PROFINET-interface_2 interfacelds[3] := Local2~PROFINET-interface_1 interfacelds[4] := Local1~PROFINET-interface_1	PLC1.X2 PLC2.X2 PLC2.X1 PLC1.X1
remotelPs	remotelPs[1].ADDR[1] := 192 remotelPs[1].ADDR[2] := 168 remotelPs[1].ADDR[3] := 1 remotelPs[1].ADDR[4] := 20 Corresponding to remotelPs[2] := 192.168.1.21 remotelPs[3] := 192.168.0.20 remotelPs[4] := 192.168.0.21	
connId	1	
port	2000	
connectionCount	4	
activeEstablished	False	
testInterval	T#50ms	

3.3.2. LComRed_ISOonTCP - Redundant connection between S7-1500H system and CPU 1518



Device properties active connection partner

Parameters	Value	Note
PLC	CPU 1517H-4 PN	6ES7517-4HQ10-0AB0
IP address interface X1	PLC1: 192.168.0.20 PLC2: 192.168.0.21	
IP address interface X2	PLC1: 192.168.1.20 PLC2: 192.168.1.21	
Minimum cycle time	1 ms	
Calling organization block	OB30	Cycle clock: 10 ms
LComRed block	LComRed_ISOonTCP	From LComRed

Device properties passive connection partner

Parameters	Value	Note
PLC	CPU 1518-4 PN/DP	6ES7 518-3AT10-0AB0
IP address interface X2	PLC1: 192.168.1.80	
IP address interface X3	PLC1: 192.168.0.80	
Minimum cycle time	1 ms	
Calling organization block	OB30	Cycle clock: 10 ms
LComRed block	LComRed_ISOonTCP	From LComRed

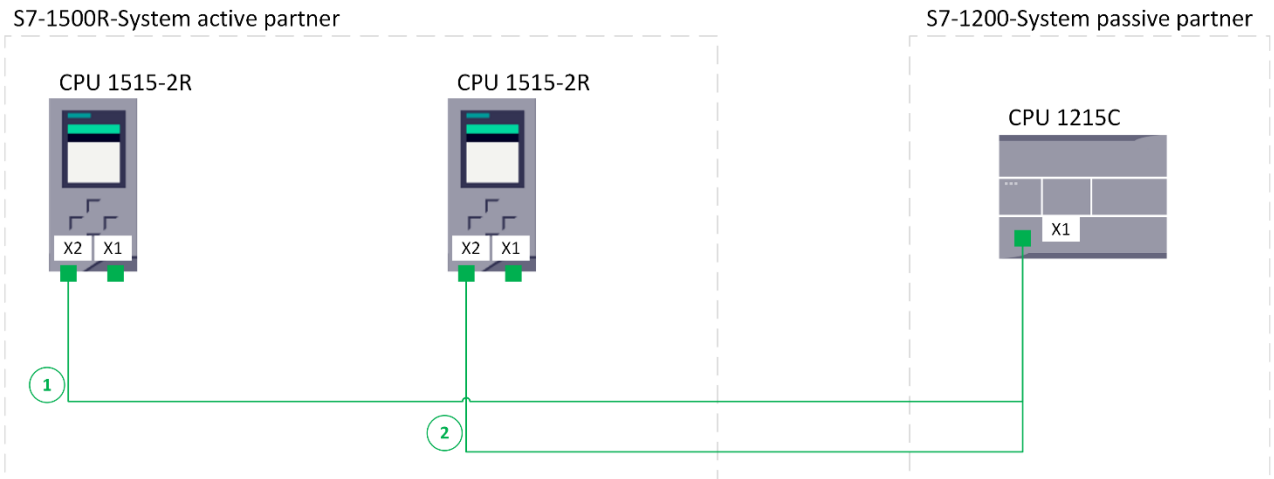
LComRed_ISOonTCP input parameter active connection partner

Parameters	Value	Note
interfacelds	interfacelds[1] := Local1~PROFINET-interface_2	PLC1.X2
	interfacelds[2] := Local2~PROFINET-interface_2	PLC2.X2
	interfacelds[3] := Local1~PROFINET-interface_1	PLC1.X1
	interfacelds[4] := Local2~PROFINET-interface_1	PLC2.X1
remotelPs	remotelPs[1].ADDR[1] := 192 remotelPs[1].ADDR[2] := 168 remotelPs[1].ADDR[3] := 1 remotelPs[1].ADDR[4] := 80	
	Corresponding to remotelPs[2] := 192.168.1.80 remotelPs[3] := 192.168.0.80 remotelPs[4] := 192.168.0.80	
connId	80	
localTSelector	HtoStd_A	
remoteTSelector	HtoStd_P	
connectionCount	4	
activeEstablished	True	
testInterval	T#50ms	

LComRed_ISOonTCP input parameter passive connection partner

Parameters	Value	Note
interfacelds	interfacelds[1] := Local~PROFINET-interface_2	PLC.X2
	interfacelds[2] := Local~PROFINET-interface_2	PLC.X2
	interfacelds[3] := Local~PROFINET-interface_GBIT_3	PLC.X3
	interfacelds[4] := Local~PROFINET-interface_GBIT_3	PLC.X3
remotelPs	remotelPs[1].ADDR[1] := 192 remotelPs[1].ADDR[2] := 168 remotelPs[1].ADDR[3] := 1 remotelPs[1].ADDR[4] := 20	
	Corresponding to remotelPs[2] := 192.168.1.21 remotelPs[3] := 192.168.0.20 remotelPs[4] := 192.168.0.21	
connId	80	
localTSelector	HtoStd_P	
remoteTSelector	HtoStd_A	
connectionCount	4	
activeEstablished	False	
testInterval	T#50ms	

3.3.3. LComRed_TCP - Redundant connection between 1500R system and S7-1215C



Device properties active connection partner

Parameters	Value	Note
PLC	CPU 1515R-2 PN	6ES7515-2RN03-0AB0
IP address interface X2	PLC1: 192.168.1.120 PLC2: 192.168.1.121	
Minimum cycle time	5 ms	
Calling organization block	OB1	
LComRed block	LComRed_TCP	From LComRed

Device properties passive connection partner

Parameters	Value	Note
PLC	CPU 1215C	6ES7 215-1AG40-0XB0
IP address interface X1	PLC1: 192.168.1.130	
Minimum cycle time	Disabled	
Calling organization block	OB30	Cycle clock: 20 ms
LComRed block	LComRed_TCP	From LComRed

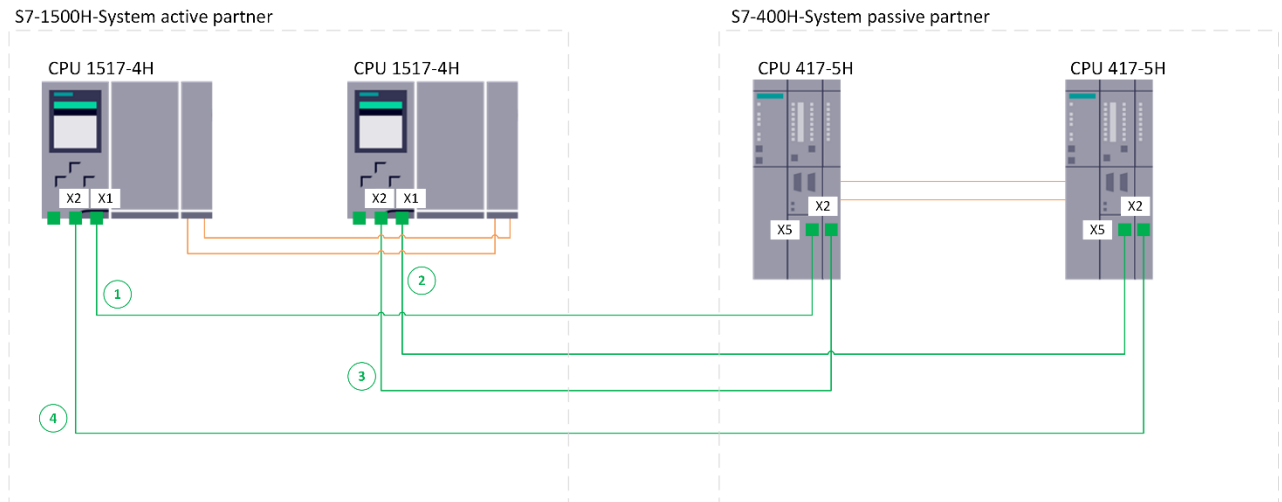
LComRed_TCP input parameter active connection partner

Parameters	Value	Note
interfacelds	interfacelds[1] := Local1~PROFINET-interface_2 interfacelds[2] := Local2~PROFINET-interface_2	PLC1.X2 PLC2.X2
remotelPs	remotelPs[1].ADDR[1] := 192 remotelPs[1].ADDR[2] := 168 remotelPs[1].ADDR[3] := 1 remotelPs[1].ADDR[4] := 130 Corresponding to remotelPs[2] := 192.168.1.130	
connId	1 300	
port	2130	
connectionCount	2	
activeEstablished	True	
testInterval	T#250ms	

LComRed_TCP input parameter passive connection partner

Parameters	Value	Note
interfacelds	interfacelds[1] := Local~PROFINET-interface_1 interfacelds[2] := Local~PROFINET-interface_1	PLC.X1 PLC.X1
remotelPs	remotelPs[1].ADDR[1] := 192 remotelPs[1].ADDR[2] := 168 remotelPs[1].ADDR[3] := 1 remotelPs[1].ADDR[4] := 120 Corresponding to remotelPs[2] := 192.168.1.121	
id	1 300	
port	2130	
connectionCount	2	
activeEstablished	False	
testInterval	T#250ms	

3.3.4. LComRed_ISOonTCP Classic - Redundant connection between 1500H-system and 400H-system



Device properties active connection partner

Parameters	Value	Note
PLC	CPU 1517H-4 PN	6ES7517-4HQ10-0AB0
IP address interface X1	PLC1: 192.168.0.20 PLC2: 192.168.0.21	
IP address interface X2	PLC1: 192.168.1.20 PLC2: 192.168.1.21	
Minimum cycle time	1 ms	
Calling organization block	OB30	Cycle clock: 10 ms
LComRed block	LComRed_ISOonTCP_Classic	From LComRed

Device properties passive connection partner

Parameters	Value	Note
PLC	CPU 417H-5H PN/DP	6ES7 417-5HT06-0AB0
IP address interface X5	PLC1: 192.168.0.50 PLC2: 192.168.0.51	
IP address interface X2 of the CP443-1	CP1: 192.168.1.50 CP2: 192.168.1.51	
Minimum cycle time	2 ms	
Calling organization block	OB1	
LComRed block	LComRed_ISOonTCP	From LComRedClassic

LComRed_ISOonTCP_Classic input parameter passive connection partner

Parameters	Value	Note
interfacelds	interfacelds[1] := Local1~PROFINET-Schnittstelle_1 interfacelds[2] := Local2~PROFINET-Schnittstelle_1 interfacelds[3] := Local2~PROFINET-Schnittstelle_2 interfacelds[4] := Local1~PROFINET-Schnittstelle_2	PLC1.X1 PLC2.X1 PLC2.X2 PLC1.X2
remotelPs	remotelPs[1].ADDR[1] := 192 remotelPs[1].ADDR[2] := 168 remotelPs[1].ADDR[3] := 0 remotelPs[1].ADDR[4] := 50 Corresponding to remotelPs[2] := 192.168.0.51 remotelPs[3] := 192.168.1.50 remotelPs[4] := 192.168.1.51	
connId	1240	
localTSelector	TSelLength[1] := 8 TSel[1, 1..8] := 35.34.49.53.4F.41.5F.31 TSelLength[2] := 8 TSel[2, 1..8] := 35.34.49.53.4F.41.5F.32 TSelLength[3] := 8 TSel[3, 1..8] := 35.34.49.53.4F.41.5F.33 TSelLength[4] := 8 TSel[4, 1..8] := 35.34.49.53.4F.41.5F.34	"54ISOA_1" "54ISOA_2" "54ISOA_3" "54ISOA_4"
remoteTSelector	TSelLength[1] := 8 TSel[1, 1..8] := 34.35.49.53.4F.50.5F.31 TSelLength[2] := 8 TSel[2, 1..8] := 34.35.49.53.4F.50.5F.32 TSelLength[3] := 10 TSel[3, 1..10] := E0.02.34.35.49.53.4F.50.5F.33 TSelLength[4] := 10 TSel[4, 1..10] := E0.22.34.35.49.53.4F.50.5F.34	"45ISOP_1" "45ISOP_2"
connectionCount	4	
activeEstablished	true	
testInterval	T#100ms	

LComRed_ISOonTCP input parameter passive connection partner

Parameters	Value	Note
interfacelds	interfacelds[1] := 16#05 interfacelds[2] := 16#15 interfacelds[3] := 16#00 interfacelds[4] := 16#10	PLC0.X5 PLC1.X5 PLC0.CP.X2 PLC1.CP.X2
remoteIPs	remoteIPs[1,1] := 16#C0 remoteIPs[1,2] := 16#A8 remoteIPs[1,3] := 16#00 remoteIPs[1,4] := 16#14 Corresponding to remoteIPs[2] := C0.A8.00.15 remoteIPs[3] := C0.A8.01.15 remoteIPs[4] := C0.A8.01.14	192 168 0 20 192.168.0.21 192.168.1.21 192.168.1.20
connId	1240	
localTSelectorLength	localTSelectorLength[1] := 16#08 localTSelectorLength[2] := 16#08 localTSelectorLength[3] := 16#0A localTSelectorLength[4] := 16#0A	
localTSelector	localTSelector[1, 1..8] := 34.35.49.53.4F.50.5F.31 localTSelector[2, 1..8] := 34.35.49.53.4F.50.5F.32 localTSelector[3, 1..10] := E0.02.34.35.49.53.4F.50.5F.33 localTSelector[4, 1..10] := E0.22.34.35.49.53.4F.50.5F.34	"45ISOP_1" "45ISOP_2"
remoteTSelectorLength	remoteTSelectorLength[1] := 16#08 remoteTSelectorLength[2] := 16#08 remoteTSelectorLength[3] := 16#08 remoteTSelectorLength[4] := 16#08	
remoteTSelector	remoteTSelector[1, 1..8] := 35.34.49.53.4F.41.5F.31 remoteTSelector[2, 1..8] := 35.34.49.53.4F.41.5F.32 remoteTSelector[3, 1..8] := 35.34.49.53.4F.41.5F.33 remoteTSelector[4, 1..8] := 35.34.49.53.4F.41.5F.34	"54ISOA_1" "54ISOA_2" "54ISOA_3" "54ISOA_4"
nextStaddr	nextStaddr[1] := 16#00 nextStaddr[2] := 16#00 nextStaddr[3] := 16#04 nextStaddr[4] := 16#24	Rack 0, Slot 4 Rack 1, Slot 4
nextStaddrLen	nextStaddrLen[1] := 16#00 nextStaddrLen[2] := 16#00 nextStaddrLen[3] := 16#01 nextStaddrLen[4] := 16#01	
connectionCount	4	
activeEstablished	false	
testInterval	T#100ms	

4. Appendix

4.1. Service and support

SiePortal

The integrated platform for product selection, purchasing and support - and connection of Industry Mall and Online support. The SiePortal home page replaces the previous home pages of the Industry Mall and the Online Support Portal (SIOS) and combines them.

- **Products & Services**
In Products & Services, you can find all our offerings as previously available in Mall Catalog.
- **Support**
In Support, you can find all information helpful for resolving technical issues with our products.
- **mySieportal**
mySiePortal collects all your personal data and processes, from your account to current orders, service requests and more. You can only see the full range of functions here after you have logged in.

You can access SiePortal via this address: sieportal.siemens.com

Technical Support

The Technical Support of Siemens Industry provides you fast and competent support regarding all technical queries with numerous tailor-made offers – ranging from basic support to individual support contracts.

Please send queries to Technical Support via Web form: support.industry.siemens.com/cs/my/src

SITRAIN – Digital Industry Academy

We support you with our globally available training courses for industry with practical experience, innovative learning methods and a concept that's tailored to the customer's specific needs.

For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page: siemens.com/sitrain

Industry Online Support app

You will receive optimum support wherever you are with the "Industry Online Support" app. The app is available for iOS and Android:



4.2. Links and literature

No.	Topic
\1\	SiePortal https://sieportal.siemens.com/en-ww
\2\	Link to the entry page of the application example https://support.industry.siemens.com/cs/ww/en/view/109763719
\3\	Link to TIA Online Documentation S7-1500/ET 200MP Manual Collection – Secure OUC docs.tia.siemens.cloud

4.3. Change documentation

Version	Date	Change
V1.0	02/2019	First version
V1.1	08/2019	Expansion with the Classic communication • LComRedClassic library for STEP 7 as of V5.5 with LComRed_ISOonTCP block • Expansion of the LComRed-V15.1_LIB with LComRed_ISOonTCP block in S7-300/400 Expansion of the LComRed-V15.1_LIB with LComRed_ISOonTCP_Classic block in S7-1200 & S7-1500
V1.2	10/2025	Feature enhancements • Secure communication for TCP communication (LComRed_TCP) • Option to check the connection of the passive connections (all function blocks) Release of all function blocks without know-how protection Adaptation of the entire library to the programming style guide for S7-1200 / S7-1500